

Week 6 Tutorial

- Change format to html if you can't render to pdf
- Change eval to true so the code runs!

Introduction

This workshop will help you familiarize yourself with uni- and multi-factor CAPM modeling.

Question 1

Consider again the stock prices for Microsoft (MSFT) and Walmart (WMT) from Jan 1990 to Mar 2024, as well as data on the risk free rate and excess returns on the “market portfolio”, denoted by MKT-RF. These are given in [XSreturns.xlsx](#).

Use these data to answer the following questions.

1. Using the data on the risk free rate, run the code below to construct excess returns on Microsoft (**xsm**) and Walmart (**xsw**). *NOTE: The RF series is in percent so we need to adjust the log-returns so that the series have similar magnitudes.*

```
library(xts)
```

```
Loading required package: zoo
```

```
Attaching package: 'zoo'
```

```
The following objects are masked from 'package:base':
```

```
as.Date, as.Date.numeric
```

```

df <- readxl::read_excel("XSreturns.xlsx")

# ignore the date column
df = df[,-1]

# Jan 1990 to March 2024

# create an xts object, but create the date first.
idx <- seq(from = as.Date("1990-01-01"), by = "month", length.out = nrow(df))
dataset.xts <- xts(df, order.by = idx)

# Check that it works - comment out when rendering
head(dataset.xts)

```

	MSFT	WMT	MKT_RF	RF
1990-01-01	0.642361	1.093507	-7.85	0.57
1990-02-01	0.685764	1.128781	1.11	0.57
1990-03-01	0.769097	1.212157	1.83	0.64
1990-04-01	0.805556	1.274974	-3.36	0.69
1990-05-01	1.013889	1.448397	8.42	0.68
1990-06-01	1.055556	1.602550	-1.09	0.63

```

# just display month-year without date
index(dataset.xts) <- as.yearmon(index(dataset.xts)) # match by year-month

# Check that it works - comment out when rendering
head(dataset.xts)

```

	MSFT	WMT	MKT_RF	RF
Jan 1990	0.642361	1.093507	-7.85	0.57
Feb 1990	0.685764	1.128781	1.11	0.57
Mar 1990	0.769097	1.212157	1.83	0.64
Apr 1990	0.805556	1.274974	-3.36	0.69
May 1990	1.013889	1.448397	8.42	0.68
Jun 1990	1.055556	1.602550	-1.09	0.63

```

# excess log return msft (%)
msft.logret <- diff(log(dataset.xts$MSFT))*100
msft.logret <- na.omit(msft.logret)# removes the first NA created by diff()
xs.msft <- msft.logret - dataset.xts$RF

# excess log return WMT (%)
wmt.logret <- diff(log(dataset.xts$WMT))*100
wmt.logret <- na.omit(wmt.logret) # removes the first NA created by diff()

```

```

xs.wmt <- wmt.logret - dataset.xts$RF

# merge 2 variables
xs.logret <- merge(xs.msft, xs.wmt)
colnames(xs.logret) <- c("xs.msft", "xs.wmt") # name the column

# Check that it works - comment out when rendering
head(xs.logret)

```

	xs.msft	xs.wmt
Feb 1990	5.968309	2.60483268
Mar 1990	10.828355	6.48631278
Apr 1990	3.941562	4.36243690
May 1990	22.321599	12.07316414
Jun 1990	3.397421	9.48386829
Jul 1990	-14.033193	0.03715025

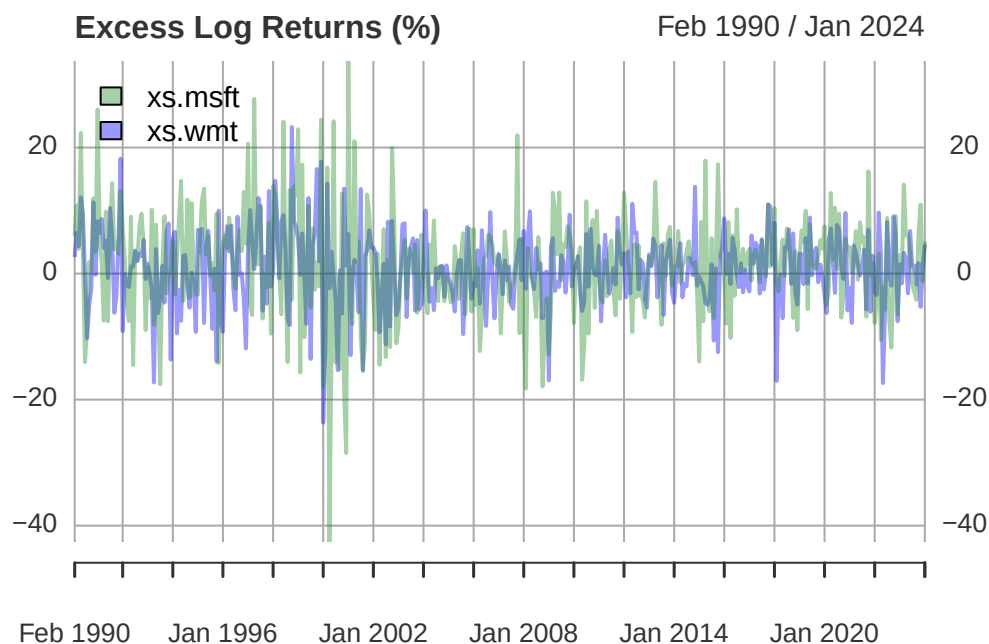
2. Use the code below to plot the two series. Which do you prefer? Can you identify any issues?

```

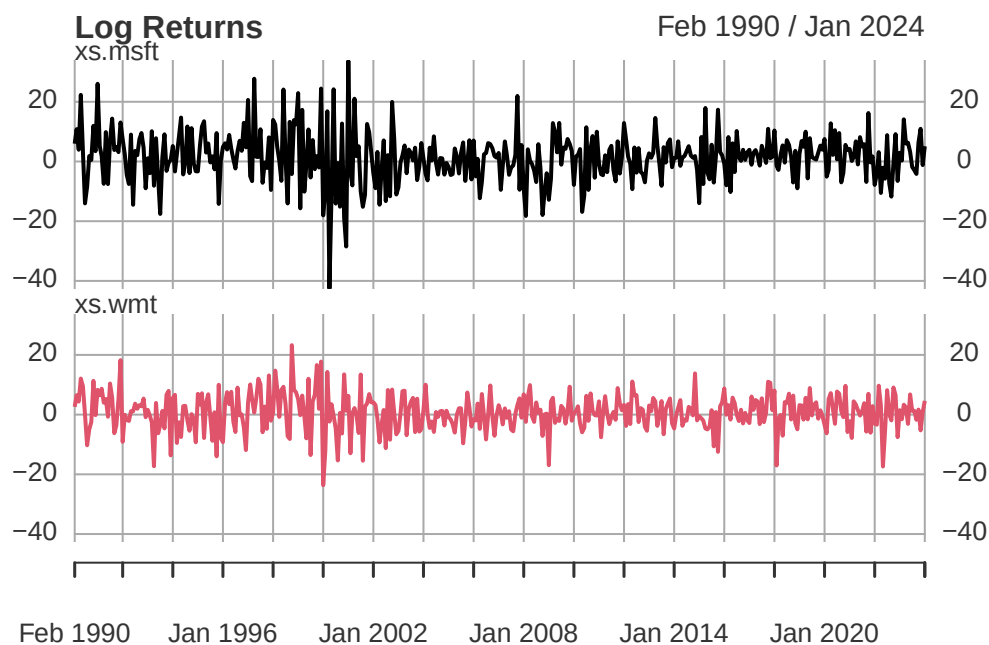
plot(xs.logret, main="Excess Log Returns (%)", legend.loc="topleft",

col = c(
  adjustcolor("forestgreen", alpha.f = 0.4),
  adjustcolor("blue", alpha.f = 0.4)
),
lwd = 2)

```



```
# Optional: create multi-panel plots
plot(xs.logret, main="Log Returns", multi.panel=TRUE)
```



```
# excess return on market portfolio MKT-RF, remove the first observation
mkt_rf <- dataset.xts$MKT_RF[-1,]
```

Answer

3. Estimate the CAPM regression model

$$r_{it} - r_{ft} = a_i + \beta_i MKT - RF_t + \epsilon_{it},$$

where $r_{it} - r_{ft}$ corresponds to excess returns on $i = M, W$ where $M = \text{Microsoft}$ and $W = \text{Walmart}$ from part 1. Interpret the estimated beta coefficient in each case.

Below is the code for the msft regression. Copy the code for wlmt (*Hint: highlight the code, CTRL+f then find and replace msft with wmt.*)

```
msft.capm.fit <- lm(xs.msft ~ mkt_rf)
summary(msft.capm.fit)
```

Call:

```
lm(formula = xs.msft ~ mkt_rf)
```

Residuals:

Min	1Q	Median	3Q	Max
-35.938	-3.674	-0.113	3.547	29.606

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.55298	0.34527	1.602	0.11
mkt_rf	1.11914	0.07696	14.541	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.883 on 406 degrees of freedom

Multiple R-squared: 0.3424, Adjusted R-squared: 0.3408

F-statistic: 211.4 on 1 and 406 DF, p-value: < 2.2e-16

```
# do t-test and compute p-value
b <- msft.capm.fit$coefficients[2]
se <- sqrt(vcov(msft.capm.fit)["mkt_rf", "mkt_rf"])

bhat <- coef(msft.capm.fit)["mkt_rf"]
se <- summary(msft.capm.fit)$coefficients["mkt_rf", "Std. Error"]

msft.t_stat <- (b - 1) / se
msft.t_stat
```

```
mkt_rf  
1.547995
```

```
# compute degrees of freedom = n-2  
  
dfr <- nobs(msft.capm.fit)-2  
  
# critical value  
# tcrit <- qt(1 - alpha/2, df )  
  
# tcrit <- qt( 0.975 , dfr )  
# tcrit  
  
# p-value  
  
# pt() is the CDF (cumulative distribution function) of the  
# Student's t distribution.  
# pt(q, df) returns the probability (area) to the left of q under a t-curve.  
  
msft.p_val <- 2 * pt(-abs(msft.t_stat), dfr )  
  
# report t-test and pvalue  
c(t = msft.t_stat, p_value = msft.p_val)
```

```
      t.mkt_rf p_value.mkt_rf  
1.5479947      0.1224027
```

```
# paste code here
```

These are simple CAPM regressions. β represents the asset's risk relative to the market. If $\beta = 1$, the asset tracks the market, if it is greater than one it is aggressive, and if it is between zero and one it is conservative (there are other interpretations for the CAPM, so check the LN's). Based upon the regression values, Microsoft is an aggressive stock, while Walmart can be classed as a conservative stock.

The intercept also tells us if the asset has adequate reward for the assumed risk. Both intercepts are above zero, so it seems as though both assets provide excess returns for the assumed risk.

4. For each asset, i , individually test the restrictions $a_i = 0$ and $\beta_i = 1$. Be sure to explain in a sentence what your conclusion means in context.

Obviously we can't simply compare the intercept to zero and the slope to 1 and make statements about the asset. We need to test the hypotheses in case the differences are due to chance.

Answer (in context!)

Question 2

First we need to get the data with the code below, using the **frenchdata** library (as you will do in your assignment). There is no need to show this code in the assignment answer. All data is of the same magnitude.

1. Estimate the four-factor CAPM model for the “Consumer” portfolio (Jan 1927-Dec 2025).

We add the SMB (Small minus Big), HML (High minus Low) and MOM (Momentum) variables to the simple CAPM equation. Remember that the multifactor CAPM models can only be used on US stocks.

- **SMB** represents the spread in returns between small- and large-sized firms. If the coefficient is positive, the asset behaves more like a small-stock portfolio. If the coefficient is negative, the asset behaves more like a big-stock portfolio.
- **HML** the spread in returns between value and growth stocks. If the coefficient is positive, then the asset behaves more like a value stock or portfolio. If the coefficient is negative, the asset behaves more like a growth stock or portfolio.
- **MOM** captures returns constructed by buying stocks with high returns and selling stocks with low returns over the same period.

Run the code below to download the Fama French data.

```
library(frenchdata)

#####
# (a) "Fama/French 3 Factors" Monthly
#####

ff3 <- download_french_data("Fama/French 3 Factors")

ff3_monthly_raw <- ff3$subsets$data[[1]]

# convert to xts object
library(xts)
# if date is like 192607, 202312, etc.
ff3_monthly_raw$date <- as.yearmon(as.character(ff3_monthly_raw$date),
                                   format = "%Y%m")
ff3.xts <- xts(
  ff3_monthly_raw[, c("Mkt-RF", "SMB", "HML", "RF")] , # drop the date column
  order.by = as.Date(ff3_monthly_raw$date)           # month index
```

```

)

# Optional: Rename the columns
colnames(ff3.xts) <- c("mkt_excess","smb","hml","rf")

# convert from "1926-07-01" to "Jul 1926"
index(ff3.xts) <- as.yearmon(index(ff3.xts))

#####
# (b) "5 Industry Portfolios"
#####

# Download the dataset (exact name on Ken French's library)
ind5 <- download_french_data("5 Industry Portfolios")

# Grab the "Average Value Weighted Returns -- Monthly"
id5.df <- ind5$subsets$data[[1]]

# convert to xts
id5.df$date <- as.yearmon(as.character(id5.df$date), format = "%Y%m")
id5.xts <- xts(
  id5.df[, c("Cnsmr","Manuf","HiTec","Hlth","Other")],# drop the date column
  order.by = as.Date(id5.df$date)                    # month index
)

# convert from "1926-07-01" to "Jul 1926"
index(id5.xts) <- as.yearmon(index(id5.xts)) # match by year-month

#####
# (c) Momentum Factor (Mom)
#####

# 1) Download the dataset (exact name on Ken French's library)
mom_raw <- download_french_data("Momentum Factor (Mom)")

# 2) Grab [[1]] monthly
mom_raw.df <- mom_raw$subsets$data[[1]]

# convert to xts
mom_raw.df$date <- as.yearmon(as.character(mom_raw.df$date), format = "%Y%m")
mom.xts <- xts(
  mom_raw.df[, "Mom" ], # drop the date column
  order.by = as.Date(mom_raw.df$date)                    # month index
)

```

```
# convert from "1926-07-01" to "Jul 1926"
index(mom.xts) <- as.yearmon(index(mom.xts)) # match by year-month

#####
# January 1997 to December 2025

id5.xts <- id5.xts["1997-01/2025-12"]

ff3.xts <- ff3.xts["1997-01/2025-12"]

mom.xts <- mom.xts["1997-01/2025-12"]

# unrestricted multi-factor
ff4f.fit <- lm(id5.xts$Cnsmr~ff3.xts$rfr~ff3.xts$mkt_excess+ff3.xts$smb+
               ff3.xts$hml+mom.xts)
summary(ff4f.fit)
```

Call:

```
lm(formula = id5.xts$Cnsmr - ff3.xts$rfr ~ ff3.xts$mkt_excess +
    ff3.xts$smb + ff3.xts$hml + mom.xts)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.9443	-1.1321	0.0826	1.0188	6.3725

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.15107	0.10602	1.425	0.15508
ff3.xts\$mkt_excess	0.84740	0.02488	34.060	< 2e-16 ***
ff3.xts\$smb	-0.09592	0.03351	-2.862	0.00446 **
ff3.xts\$hml	0.07909	0.03192	2.477	0.01372 *
mom.xts	-0.01853	0.02270	-0.816	0.41490

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.933 on 343 degrees of freedom

Multiple R-squared: 0.8, Adjusted R-squared: 0.7976

F-statistic: 343 on 4 and 343 DF, p-value: < 2.2e-16

2. Test the hypothesis that $\alpha = 0$ and interpret the outcome of the test.

[Answer in context](#)

3. Conduct a hypothesis test that the SMB factor is statistically significant.

Answer in context

4. Test the hypothesis that the residuals display no serial correlation against the alternative that they are serially correlated. Also look at the ACF (save the **Correlogram.R** file into your project) of the residuals. Is the market efficient?

Refer to the [Diagnostic test handout](#). Here we are doing a test for serial correlation. It is assumed that all prior information in the market has been accounted for (that is, the EMH holds), so there should be no time dependence in the regression.

We have not been told what order of serial correlation to test for. Since the data is monthly, testing for up to order 12 is sensible.

The auxiliary regression is:

$$\begin{aligned}\hat{\epsilon}_t = & \beta_0 + \beta_1 MKT - RF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t \\ & + \gamma_0 + \gamma_1 \hat{\epsilon}_{t-1} + \gamma_2 \hat{\epsilon}_{t-2} + \dots + \gamma_{12} \hat{\epsilon}_{t-12} + v_t\end{aligned}$$

We test $H_0 : \gamma_i = 0$, for $i = 1, \dots, 12$ against H_1 : at least one $\gamma_i \neq 0$ for $i = 1, \dots, 12$.

Run the code below.

```
# change eval = TRUE to run code when you render!
# Serial Correlation LM-test
lmtest::bptest(ff4f.fit, order = 12, type = "Chisq")
```

The test Statistic is $W = nR^2$, which has a χ^2_{12} distribution under H_0 . The calculated value is ??? with a p-value of ???. We ??? and conclude that ???. The market is ???.

We can also use a correlogram to do this test informally. Use the code below to obtain the ACF of the residuals (this will use the function in the **Correlogram.R** file.

```
# change eval = TRUE to run code when you render!

# correlogram of residuals

source("Correlogram.R") # You may need to change the path here

res <- residuals(ff4f.fit) # or res <- ff4f.fit$residuals
correlogram(res, max.lag = 10, fit = ff4f.fit)
```

Answer

Extra answer

5. Conduct a test that the residuals display homoskedasticity against the alternative that they are heteroskedastic.

Here we will do White's test in R. You would need to write out the auxiliary regression, the null and alternative and the form and distribution of the test statistic under the null in an assignment. These are given in the Diagnostic test handout, and is similar to the answer above. Here we will just get the R output.

Run the code below.

```

# change eval = TRUE to run code when you render!

# white test,
# include cross products and squares
# manually compute
# I(.) is important because it tells R to treat ^2 as arithmetic
# (x1+x2)^2 gives x1, x2 and their cross product

white_aux <- lm(I(res^2) ~
                (ff3.xts$mkt_excess+ff3.xts$smb+ff3.xts$hml+mom.xts)^2 +
                I(ff3.xts$mkt_excess^2)+I(ff3.xts$smb^2)+I(ff3.xts$hml^2)+
                I(mom.xts^2))
summary(white_aux)

white.test = summary(white_aux)$r.squared * nobs(white_aux)
white.test

# This white.test has a chi-square distribution and the df equals the number
# of regressors in the auxiliary regression excluding the constant
# (i.e., the number of slope coefficients being tested jointly).
# beta1 = beta2 = ... = betak

# Find the df = k-1
dfw <- length(coef(white_aux)) - 1
dfw

# Calculate the p-value
pval <- 1 - pchisq(white.test, dfw)
pval

```

The test statistic is $W = nR^2$. This has a χ^2_{14} distribution under H_0 . The calculated value is ??? with a p-value of ???. We would ???.

This means that ???.